

Personal Contribution Report

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Project: NBP Analyzer

Team: BunniHopPon

Role: Developer

Project Membership and Role

I am a member of the BunniHopPon project group, responsible for the development of the "NBP Analyzer" system. My official role in the team is Developer. My primary focus was on implementing the core statistical logic, developing the user interface for session trends, and ensuring the application adheres to the defined Software Requirements Specification (SRS)

List of Individual Contributions

1. Core Logic and Data Fetching

- API Integration: Implemented the `fetchNbpRates` and `normalizeNbpResponse` functions to communicate with the NBP API and process the incoming JSON data.
- Statistical Modules: Developed and implemented core mathematical functions for analyzing currency data, including:
 - `calcMedian`: Calculation of the median exchange rate
 - `calcMode`: Identification of the dominant (mode) value
 - `calcStdDev`: Calculation of the standard deviation and coefficient of variation
 - `countSessionTrends`: Logic to determine the number of rising, falling, and unchanged sessions

2. UI/UX Development (Session Trends)

- Component Architecture: Initialized and structured the `SessionTrendsPanel`, providing a clean interface for currency and period selection
- Data Visualization: Integrated the `recharts` library to provide visual representation of exchange rate fluctuations through line charts
- Selector Implementation: Developed custom currency and date selectors to allow users to specify analysis parameters.

- Loading States: Implemented loading spinners and disabled interactive controls during data fetching to enhance user experience and prevent race conditions

3. Frontend Integration and Compliance

- Main App Integration: Integrated the SessionTrendsPanel into the main App component and optimized the overall layout
- SRS Adherence: Enforced project-wide styling constraints, including the light theme, blue accent colors, and full-width layout as specified in the GUI design documents
- Validation Logic: Implemented input validation to prevent the selection of future dates and invalid analysis periods, ensuring data integrity

4. Quality Assurance and Maintenance

Bug Fixing: Resolved critical application issues, including "black screen" crashes and UI rendering bugs

Refactoring: Conducted regular code refactoring and linting fixes to maintain code readability and project standards